DJEZZY NATIONAL SOC PLATFORM

100% On-Premises Hardware Architecture
& Implementation Guide

Document Classification	CONFIDENTIAL - INTERNAL USE ONLY
Deployment Model	PURE ON-PREMISES (Zero Cloud Dependencies)
Air-Gap Capability	FULLY SUPPORTED
Data Sovereignty	100% Algerian Territory Compliance
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Document Scope

This document provides comprehensive hardware architecture specifications and implementation procedures for deploying the Djezzy National Security Operations Center (SOC) Platform entirely within Djezzy-owned physical facilities. All infrastructure components including compute servers, storage systems, network fabric, backup systems, disaster recovery site, development environments, and operational tooling are deployed on-premises with zero dependencies on external cloud services or third-party hosted platforms.

Critical Design Constraint: Zero External Connectivity Required

The platform architecture supports fully isolated (air-gapped) operation mode. Once initial deployment and container image loading is complete, the SOC platform can operate indefinitely without any external network connections. This capability is mandatory for national telecommunications security infrastructure subject to Algerian data protection regulations and national security requirements.

1. Executive Summary

This Hardware Architecture Guide defines the complete physical infrastructure required to deploy and operate the Djezzy National Security Operations Center (SOC) Platform as a 100% on-premises solution with zero cloud dependencies. The platform has been architected from the ground up to support air-gapped operation, ensuring that all security monitoring, event processing, threat intelligence, incident response, and analytical capabilities function independently of external network connectivity once deployment is complete. This design aligns with stringent Algerian telecommunications regulatory requirements mandating that subscriber data, network traffic records, and security intelligence remain within national borders under complete organizational control.

The recommended deployment spans two geographically separated data center sites operating in active-passive configuration for disaster recovery purposes. The primary site houses the full production cluster comprising fourteen (14) physical servers organized into functional roles optimized for the specific workload characteristics of each service category. The secondary disaster recovery site maintains a reduced-capability standby configuration with eight (8) servers capable of assuming full production operations within the defined recovery time objective of thirty minutes following site failure declaration. Both sites operate entirely independently with no requirement for cross-site connectivity during normal operations, though dedicated fiber links enable asynchronous data replication when business continuity policies permit such connectivity.

1.1 Platform Resource Requirements

The aggregate resource requirements reflect production-scale deployment supporting fifty billion security events annually, eighty billion Call Detail Records per year, real-time monitoring for fifteen million mobile subscribers, and sustained packet capture at one hundred gigabits per second line rate across monitored network segments. These figures incorporate twenty percent headroom above measured baseline consumption to accommodate traffic spikes during security incidents, organic growth over the thirty-six month planning horizon, and capacity reserved for additional data sources that may be integrated during future expansion phases.

Resource Category	Primary Site	DR Site	Total Capacity	Recommended Config
Compute (vCPU Cores)	138 cores	72 cores	210 cores	Intel Xeon Gold/Platinum
RAM Memory	512 GB	256 GB	768 GB	DDR4-3200 ECC Registered
NVMe Primary Storage	85 TB	40 TB	125 TB	Enterprise U.3 NVMe
HDD Archive Storage	120 TB	60 TB	180 TB	Enterprise SAS 12Gbps
SSD Cache/Log Storage	8 TB	4 TB	12 TB	NVMe RAID 1 pairs
Network (Data Plane)	400 Gbps	200 Gbps	600 Gbps	100GbE/25GbE fabric
Network (Management)	40 Gbps	20 Gbps	60 Gbps	Dedicated OOB network

Table 1.1: Aggregate Resource Requirements by Site

1.2 Design Principles

- Data Sovereignty: All security data, subscriber information, network records, and threat intelligence reside exclusively on Djezzy-owned hardware within Algerian territory. No data ever traverses external networks or foreign jurisdictions under any circumstances.
- Air-Gap Operation: The platform architecture supports fully disconnected operation. Container images, software packages, and configuration artifacts are loaded during initial deployment via secure media transfer. Subsequent operation requires zero inbound or outbound network connectivity.
- High Availability: All critical services deploy in redundant configurations with automatic failover. The primary site achieves 99.99% availability through N+1 server redundancy, dual power feeds, multipath networking, and hot-swappable component replacement without service interruption.
- Disaster Recovery: Geographically separated DR site maintains standby capability for all critical functions. Recovery Point Objective (RPO) of fifteen minutes maximum; Recovery Time Objective (RTO) of thirty minutes for Tier-1 services following failover initiation.
- Performance at Scale: Infrastructure sizing supports sustained ingestion of fifty thousand events per second, concurrent query handling for fifty analysts, sub-three-second search response times on multi-billion-event indices, and line-rate packet capture without loss at 100Gbps.

2. Primary Site Architecture

The primary site constitutes the main production facility housing the complete Djezzy SOC Platform in full operational configuration. This site processes all live security events, hosts active database instances serving analyst queries, executes real-time alerting logic, performs automated threat detection, and provides the primary user interface for security operations personnel. The primary site infrastructure comprises fourteen (14) physical servers arranged across five functional clusters, interconnected via a high-performance spine-leaf switching fabric with dedicated management and out-of-band administration networks physically separated from data plane traffic.

2.1 Server Inventory - Primary Site

#	Server Role	Qty	Processor Specification	Memory	Primary Storage	Network Interfaces
1	SIEM Master Nodes (Wazuh/Elasticsearch Master)	2	Dual Intel Xeon Gold 6348 (28C/56T) 2.6GHz base, 3.5GHz turbo	512GB DDR4 3200MHz ECC RDIMM (8x64GB)	2x 960GB NVMe RAID 1 (OS) + 2x 1.92TB NVMe RAID 1 (Logs)	2x 25GbE SFP28 (Data) 1x 10GbE BASE-T (Mgmt) 1x 1GbE IPMI
2	Elasticsearch Data Nodes (Hot/Warm Tiers)	4	Dual Intel Xeon Gold 6348 (28C/56T) 2.6GHz base, 3.5GHz turbo	512GB DDR4 3200MHz ECC RDIMM (8x64GB)	8x 7.68TB NVMe U.3 RAID 10 (~27TB usable per node)	2x 25GbE SFP28 (Data) 1x 10GbE BASE-T (Mgmt) 1x 1GbE IPMI
3	Database Cluster (PostgreSQL/Kafka/Redis)	3	Dual Intel Xeon Silver 4314 (32C/64T) 2.4GHz base, 3.4GHz turbo	384GB DDR4 3200MHz ECC RDIMM (12x32GB)	4x 3.84TB NVMe RAID 10 (DB data) + 4x 3.84TB NVMe RAID 10 (Kafka logs) + 2x 960GB NVMe RAID 1 (Redis)	2x 25GbE SFP28 (Data) 1x 10GbE BASE-T (Mgmt) 1x 1GbE IPMI
4	NSM/Packet Capture (Suricata/Zeek/Arkime)	2	Dual Intel Xeon Gold 6248R (20C/40T) 3.0GHz base, 4.1GHz turbo	256GB DDR4 3200MHz ECC RDIMM (8x32GB)	2x 960GB NVMe RAID 1 (OS/Cache) + 6x 3.84TB NVMe RAID 10 (PCAP) (~10TB usable)	2x 100GbE QSFP28 (Packet Capture) 1x 25GbE SFP28 (Data Export) 1x 1GbE IPMI
5	Application Services (SOAR/ThreatIntel/Vuln/API)	2	Single Intel Xeon Gold 6230 (20C/40T) 2.1GHz base, 3.2GHz turbo	128GB DDR4 3200MHz ECC RDIMM (4x32GB)	2x 480GB NVMe RAID 1 (OS/Apps)	2x 25GbE SFP28 (Data) 1x 10GbE BASE-T (Mgmt) 1x 1GbE IPMI
6	Archive/Backup Server (Long-term Retention)	1	Dual Intel Xeon Silver 4310 (12C/24T) 2.1GHz base, 3.0GHz turbo	128GB DDR4 3200MHz ECC RDIMM (4x32GB)	2x 960GB NVMe RAID 1 (OS) + 24x 16TB Enterprise SAS RAID 6 (~300TB usable)	2x 25GbE SFP28 (Backup Network) 1x 10GbE BASE-T (Mgmt) 1x 1GbE IPMI

Table 2.1: Primary Site Complete Server Specifications

2.2 Detailed Component Analysis

2.2.1 SIEM Master Nodes (2 servers)

The SIEM master nodes serve as the coordination layer for the entire security information and event management stack. Each node runs Wazuh manager instances responsible for agent registration, rule evaluation, and alert generation across the monitored environment. Additionally, these nodes host Elasticsearch master-eligible nodes that manage cluster state, index metadata, and shard allocation decisions. The Kibana visualization platform deploys behind a load balancer distributing analyst queries across both nodes, ensuring continued dashboard availability during single-node maintenance events. Processor selection prioritizes single-thread performance critical for Elasticsearch aggregation operations and Wazuh rule engine throughput. Memory allocation provides ample heap space for Elasticsearch JVM (30GB per node), Wazuh process memory (8GB), and operating system filesystem cache to accelerate frequent index segment access.

2.2.2 Elasticsearch Data Nodes (4 servers)

Elasticsearch data nodes constitute the storage and query engine backbone of the SIEM platform, hosting the actual indexed security event data served to analyst queries. The four-node configuration provides both high availability through replica distribution and horizontal query scaling as concurrent user load increases. Each node contributes approximately twenty-seven terabytes of usable storage in RAID 10 arrays delivering over one hundred thousand random read IOPS and eighty thousand random write IOPS, sufficient to maintain sub-second query response times even during intensive investigation sessions involving complex aggregations across billions of events. Storage is partitioned into hot tier (SSD, seven-day retention with highest query performance), warm tier (SSD, thirty-day retention with slightly elevated latency), and frozen tier (automatically migrated to archive server based on index lifecycle policies). Memory configuration allocates sixty percent to Elasticsearch heap, twenty percent to operating system page cache for segment file buffering, and twenty percent reserved for other system processes and kernel structures.

2.2.3 Database Cluster Nodes (3 servers)

The database cluster provides persistent storage for structured data including security case records (TheHive), threat intelligence observations (MISP/OpenCTI), vulnerability findings (DefectDojo), subscriber profiles, CDR correlation results, and platform configuration state. PostgreSQL 16 deploys in streaming replication topology with one primary accepting writes and two hot standbies capable of immediate promotion upon primary failure. PgBouncer connection pooling sits between application containers and database instances, maximizing connection efficiency for the containerized microservices architecture generating thousands of concurrent database connections. Kafka brokers co-located on these nodes benefit from the same high-performance NVMe storage subsystem, with dedicated logical volumes isolating Kafka log segments from PostgreSQL data files to prevent I/O contention between sequential write (Kafka) and random access (PostgreSQL) patterns.

2.2.4 NSM/Packet Capture Nodes (2 servers)

Network Security Monitoring nodes represent the most performance-critical infrastructure components, responsible for capturing, parsing, and analyzing network traffic at line rate without packet loss. Each node features dual 100 Gigabit Ethernet interfaces configured in active-active bond, receiving mirrored traffic from core routerSPAN ports or network tap devices. Suricata intrusion detection engine processes packets using multi-threaded architecture distributing flow analysis across available CPU cores, while Zeek (formerly Bro) performs deep protocol parsing extracting application-layer intelligence for security-relevant protocol analysis. Captured PCAP files stream to Arkime (formerly Moloch) for indexed packet-level retrieval enabling investigators to reconstruct individual sessions, extract file transfers, and visualize communication patterns. The ten terabytes of local PCAP storage per node retains approximately forty-eight hours of full-packet capture at 50Gbps average utilization before rotation, with older captures migrated to the archive server for longer retention subject to storage capacity and investigative requirements.

2.3 Storage Architecture Details

Service Component	Storage Type	Raw Capacity	Usable (RAID)	IOPS Target	Retention Period
Elasticsearch Hot Tier	NVMe U.3 RAID 10	240 TB (4x60TB)	~108 TB	100K+ random R/W	7 days (active)
Elasticsearch Warm Tier	NVMe U.3 RAID 10	120 TB (2x60TB)	~54 TB	75K+ random R/W	30 days (frequent)
PostgreSQL Database	NVMe RAID 10	46 TB (3x15.4TB)	~20 TB	25K+ random R/W	Online (permanent)
Kafka Event Log	NVMe RAID 10	46 TB (3x15.4TB)	~20 TB	75K+ seq write	14 days (configurable)
Redis Cache Layer	NVMe RAID 1	6 TB (3x2TB)	~3 TB	200K+ random R/W	Volatile (cache)
PCAP Capture (Active)	NVMe RAID 10	120 TB (2x60TB)	~54 TB	50K+ seq write	48 hours (hot)
PCAP Archive	SAS HDD RAID 6	384 TB (24x16TB)	~300 TB	2K+ seq write	90 days (extended)
OS/System Volumes	NVMe RAID 1	28 TB (14x2TB)	~14 TB	Standard	N/A
Backup Repository	SAS HDD RAID 6	384 TB (dedicated)	~300 TB	2K+ seq write	365 days (rotating)

Table 2.2: Complete Storage Allocation Matrix

3. Disaster Recovery Site Architecture

The disaster recovery (DR) site provides geographic redundancy ensuring business continuity when the primary site becomes unavailable due to natural disasters, facility-wide power failures, extended cooling outages, network provider disruptions, or other catastrophic events affecting the primary location. The DR site operates in hot-standby configuration with all services pre-deployed and synchronized via asynchronous replication, enabling rapid failover without requiring software installation, configuration, or data restoration procedures during emergency conditions. Site separation distance exceeds one hundred kilometers to provide meaningful protection against regional disasters while remaining within Algeria's national boundaries to satisfy data sovereignty requirements.

3.1 DR Site Server Inventory

#	Server Role	Qty	Processor	Memory	Storage	Purpose
1	DR SIEM/Analytics	2	Dual Xeon Gold 5318Y (24C/48T)	256GB DDR4	4x 3.84TB NVMe RAID 10 + 2x 960GB OS	Standby ES/Kibana/ Wazuh managers
2	DR Database	2	Dual Xeon Silver 4314 (32C/64T)	192GB DDR4	4x 3.84TB NVMe RAID 10	Standby PG/Kafka replicas
3	DR Application	2	Single Xeon 6230 (20C/40T)	64GB DDR4	2x 480GB NVMe RAID 1	Standby SOAR/Threat Intel/Vuln apps
4	DR Archive/Restore	1	Dual Xeon 4310 (12C/24T)	96GB DDR4	12x 16TB SAS RAID 6 + 2x 960GB NVMe	Replicated archive restore target
5	DR Management/Gateway	1	Single Xeon 4314 (16C/32T)	32GB DDR4	2x 240GB SSD RAID 1	VPN concentrator, bastion host, monitoring

Table 3.1: Disaster Recovery Site Server Specifications

3.2 Replication Architecture

Data synchronization between primary and DR sites employs multiple replication mechanisms tailored to the consistency and latency requirements of each data type. For the Elasticsearch security event indices, Cross-Cluster Replication (CCR) continuously pulls new documents from the primary cluster to the DR cluster with configurable follower lag tolerance typically set to fifteen seconds maximum under normal network conditions. PostgreSQL database replication utilizes asynchronous streaming replication with WAL shipping, achieving RPO of approximately five seconds when inter-site connectivity is operational. Kafka topics replicate via MirrorMaker 2 (or native Kafka ReplicaLink in newer versions) maintaining topic offset synchronization that enables consumer groups to resume processing at the correct position following failover with minimal message duplication or loss.

Data Type	Replication Method	RPO Target	Bandwidth Req	Failover Action

Security Events (ES)	Cross-Cluster Replication	< 15 min	~500 Mbps avg	Promote follower index, redirect queries
PostgreSQL Database	Streaming Replication (async)	< 5 min	~100 Mbps avg	pg_ctl promote, update DNS/VIP
Kafka Event Topics	MirrorMaker 2 / ReplicaLink	< 10 min	~200 Mbps avg	Consumer offset sync, restart consumers
Configuration (GitOps)	Git repository push mirror	< 1 min	< 1 Mbps	Git pull, apply configs
Container Images	Registry push replication	< 30 min	~50 Mbps (initial)	Local registry already synced
PCAP Archives	Async rsync / object sync	< 1 hour	~1 Gbps peak	Accept gap, resume capture
Secrets/Credentials	Vault replication / HSM sync	< 5 min	< 1 Mbps	Standby Vault unseal

Table 3.2: Inter-Site Replication Configuration

3.3 Failover Procedures

Disaster recovery failover follows documented runbook procedures executed by trained operations personnel (or automatically via orchestrated workflows if implemented). Upon detection of primary site failure exceeding predefined thresholds (typically five minutes of complete unreachability for critical services), the incident commander declares disaster recovery activation. Automated health check scripts verify DR site service status and data currency, confirming replication lag falls within acceptable bounds for each critical data store. Database promotion executes via pre-configured automation promoting the warm standby to primary role, while DNS updates (or VIP migration via anycast routing) redirect client traffic to the DR site IP addresses. Target failover time from declaration to full service availability is thirty minutes for Tier-1 services (event ingestion, alerting, basic search) and sixty minutes for Tier-2 services (full analytical capabilities, historical search, case management).

- Verify primary site outage through multiple independent confirmation channels
- Activate DR site operations room and assemble response team (virtual or physical)
- Execute automated health verification scripts confirming DR service readiness
- Review replication lag reports for all critical data stores against RPO targets
- Obtain formal authorization from designated incident commander for failover execution
- Execute database promotion sequence (PostgreSQL promote, ES index promotion)
- Update DNS records or migrate anycast VIPs to point to DR site addresses
- Validate end-to-end functionality through synthetic transaction testing
- Communicate status update to stakeholders including degraded capability notices
- Monitor DR site performance closely during initial hours post-failover
- Document timeline and lessons learned for post-incident review

4. Network Infrastructure Design

Network infrastructure for the Djezzy SOC Platform implements a defense-in-depth architecture combining physical segmentation, logical isolation, encrypted communications, and comprehensive monitoring visibility. The design accommodates the extreme throughput requirements of line-rate packet capture while providing robust connectivity between containerized microservices with predictable latency characteristics essential for real-time security analytics. All network equipment resides within Djezzy facilities with no dependency on external connectivity providers for internal platform operation.

4.1 Network Segmentation Zones

Zone Name	CIDR Range	VLAN ID	Purpose	Connected Components	Security Level
soc-management	172.16.0.0/16	VLAN 100	OOB admin, IPMI, device management	Switch mgmt, IPMI, Bastion hosts	CRITICAL - Isolated
soc-frontend	172.28.0.0/16	VLAN 200	User-facing apps, API gateway, dashboards	Next.js, Kong GW, Grafana, Analyst VPN	RESTRICTED - Auth required
soc-backend	172.29.0.0/16	VLAN 300	Core services, SOAR, Threat Intel	TheHive, Cortex, MISP, OpenCTI, DefectDojo	INTERNAL - Service-to-service
soc-events	172.30.0.0/16	VLAN 400	Event processing, SIEM, Analytics	Wazuh, ES, Kafka, PostgreSQL, Redis	SENSITIVE - High throughput
soc-capture	172.31.0.0/16	VLAN 500	Packet capture, NSM engines	Suricata, Zeek, Arkime, TAPs	RESTRICTED - No direct access
soc-backup	172.32.0.0/16	VLAN 600	Backup traffic, archive replication	Backup server, DR link, NAS/SAN arrays	INTERNAL - Bulk transfer
soc-monitoring	172.33.0.0/16	VLAN 700	Observability, metrics, logging	Prometheus, AlertManager, Log aggregation	INTERNAL - Ops team only

Table 4.1: Network Zone Segmentation Scheme

4.2 Switching Fabric Topology

The primary site network employs spine-leaf architecture providing non-blocking bisectional bandwidth between any pair of endpoints while minimizing hop count and deterministic latency paths. Two spine switches form the aggregation layer connecting all leaf switches in full-mesh topology with 100 Gigabit Ethernet links. Leaf switches provide server-facing ports at appropriate speeds matching each server role's requirements: 100GbE for NSM capture nodes, 25GbE for Elasticsearch and database nodes, and 10GbE for management and application servers. This hierarchical design simplifies operations by eliminating spanning tree protocols (using ECMP routing instead), provides easy capacity addition through leaf spin-up, and contains failure domains to individual leaf switches affecting only directly attached servers rather than causing fabric-wide disruption.

Device Role	Model Example	Port Configuration	Uplink/Downlink	Quantity	Redundancy

Spine Switch	Cisco Nexus 9504 w/ 9700-48X linecard	48x 100GbE QSFP28 + 6x 400GbE QSFP-DD	Spine-Spine: 400GbE Spine-Leaf: 100GbE	2	N+1 (one spare)
Leaf (Capture)	Cisco Nexus 9364CD Arista 7280R3	48x 25GbE SFP28 + 8x 100GbE QSFP28	Down: 25/100GbE Up: 100GbE to Spine	2	Paired (A/B)
Leaf (Compute)	Cisco Nexus 9364C Arista 7050X3	48x 25GbE SFP28 + 8x 100GbE QSFP28	Down: 25GbE Up: 100GbE to Spine	2	Paired (A/B)
Leaf (Storage)	Cisco Nexus 9364C Arista 7050X3	48x 25GbE SFP28 + 8x 100GbE QSFP28	Down: 25GbE Up: 100GbE to Spine	1	N+1 (spare port)
Management	Cisco Catalyst 9300-48T Juniper EX4300-48T	48x 1GbE/10GbE BASE-T + 4x 10GbE SFP+	Down: 1/10GbE Up: 10GbE to Mgmt Core	2	N+1 redundant
Out-of-Band	Cisco Catalyst 9200 Lantronix SLC8000	48x 1GbE Serial console	IPMI/Console only	1	Standalone OK

Table 4.2: Network Switch Inventory and Specifications

4.3 Inter-Site Connectivity

Connectivity between primary and disaster recovery sites utilizes diverse path fiber circuits provisioned from two different telecommunications carriers to eliminate single point of failure in the transport layer. Primary circuit provides ten Gigabit Ethernet bandwidth sufficient for continuous asynchronous replication of all critical data stores under normal operation. Secondary circuit offers two Gigabit Ethernet capacity serving as automatic failover path should the primary circuit degrade or fail. Both circuits terminate on carrier-grade routers at each site running BGP for dynamic path selection, with encryption applied via IPsec tunnels protecting all inter-site traffic confidentiality and integrity. During normal operations, replication traffic consumes approximately six to eight Gigabits of the primary circuit capacity, leaving headroom for administrative access and emergency bulk transfers. Circuit diversity ensures that fiber cuts, carrier maintenance windows, or provider outages affect only one path without disrupting replication continuity.

Circuit Attribute	Primary Link	Secondary Link
Provider	Algerie Telecom (example)	Djezzi (alternative carrier)
Bandwidth	10 Gbps Ethernet	2 Gbps Ethernet
Technology	DWDM dark fiber / MPLS	Dedicated leased line
Latency (RTD)	< 5ms (< 150km path)	< 8ms (diverse route)
Encryption	AES-256 IPsec tunnel	AES-256 IPsec tunnel
Routing Protocol	BGP with local pref	BGP (lower preference)
Primary Use	DB/ES/Kafka replication	Admin access, backup overflow
SLA Availability	99.99% (52 min/yr downtime)	99.9% (8.7 hr/yr downtime)

Table 4.3: Inter-Site WAN Circuit Specifications

5. Physical Facility Requirements

Successful operation of the Djezzy SOC Platform requires physical facility infrastructure meeting specific power, cooling, space, and environmental standards. This section outlines requirements for both primary and disaster recovery sites, providing guidance for facility managers and data center planners preparing to host the platform hardware. Organizations with existing data center facilities should assess current capabilities against these requirements to identify necessary upgrades or expansions prior to equipment installation.

5.1 Power Requirements

Component Category	Quantity	Avg Draw (kW)	Peak Draw (kW)	Total Avg (kW)	Total Peak (kW)
SIEM Master Servers	2	0.8 kW	1.2 kW	1.6 kW	2.4 kW
Elasticsearch Data Nodes	4	1.0 kW	1.5 kW	4.0 kW	6.0 kW
Database Cluster Nodes	3	0.9 kW	1.4 kW	2.7 kW	4.2 kW
NSM/Capture Nodes	2	1.2 kW	1.8 kW	2.4 kW	3.6 kW
Application Servers	2	0.5 kW	0.8 kW	1.0 kW	1.6 kW
Archive/Backup Server	1	0.7 kW	1.1 kW	0.7 kW	1.1 kW
Network Switches (all)	12	0.3 kW	0.5 kW	3.6 kW	6.0 kW
Storage Arrays (SAN/NAS)	2	1.5 kW	2.5 kW	3.0 kW	5.0 kW
UPS Systems (overhead)	-	-	-	2.0 kW	3.0 kW
Cooling (PUE factor 1.5)	-	-	-	21.0 kW	33.0 kW
TOTAL PRIMARY SITE	-	-	-	~42 kW	~66 kW

Table 5.1: Primary Site Power Budget Estimate

Power infrastructure must deliver the calculated peak capacity with N+1 redundancy at minimum, meaning UPS and power distribution systems should be rated for at least seventy-five kilowatts to accommodate the sixty-six kilowatt peak load with headroom for transient spikes and future growth. Dual-feed configurations connect each server to separate power distribution units (PDUs) fed from independent UPS systems, themselves connected to separate utility feeds or generator sources where available. Automatic transfer switches (ATS) handle failover between utility and generator power within ten milliseconds, preventing server reboot during power source transitions. Battery runtime should sustain full load for minimum fifteen minutes allowing orderly graceful shutdown or generator start-up completion for extended outages.

5.2 Cooling Requirements

Cooling infrastructure must remove approximately forty-two kilowatts of continuous heat load (sixty-six kilowatts peak) from the primary site equipment footprint while maintaining ASHRAE-recommended environmental parameters for IT equipment operation. Recommended supply air temperature ranges from eighteen to twenty-seven degrees Celsius with relative humidity between forty and sixty percent non-condensing. Hot aisle/cold aisle containment strongly recommended to prevent recirculation mixing and improve cooling efficiency. Precision air conditioning units (CRAC or CRAH) with variable speed fans and economizer modes reduce energy consumption during cooler ambient periods. Redundant cooling units (N+1 configuration) ensure continued operation during single-unit maintenance or failure scenarios.

5.3 Space Requirements

Item	Quantity	RU Height	Footprint (sq meters)	Notes
Server (2U each)	14 servers	28 RU	2.8 sq m	Standard 19" rack mount
Switches (1U each)	12 switches	12 RU	1.2 sq m	Includes spares
Storage Arrays	2 units	14 RU	1.4 sq m	SAN/NAS enclosures
UPS Systems	2 units	10 RU	2.0 sq m	Battery cabinets adjacent
PDU/Power	8 units	4 RU	-Mounted	Zero-U or 1U PDUs
Patch Panels/Cabling	-	8 RU	-Mounted	Fiber/copper management
Cable Management	-	4 RU	-Mounted	Horizontal/vertical managers
Total Equipment	-	78 RU	7.4 sq m	Excluding clearance
Recommended Rack Count	4 racks (42U each)	168 RU total	12 sq m incl. clearance	~50% utilization for airflow

Table 5.2: Physical Space Requirements

5.4 Physical Security Requirements

Given the sensitive nature of security operations data and the national critical infrastructure context, physical security controls must meet or exceed industry best practices for data center facilities. Access control requires multi-factor authentication (badge + PIN/biometric) at building entry, with additional badge authentication at computer room entrance. CCTV surveillance with ninety-day retention covers all entry points, aisles, and equipment rows. Visitor escort mandatory for all non-authorized personnel. Environmental monitoring sensors detect water leaks, smoke, temperature excursions, and humidity anomalies with automated alerting to operations center and facilities management. Fire suppression uses clean agent systems (FM-200, Novec 1230, or inert gas) suitable for electronic equipment protection without water damage risk.

6. Implementation Roadmap

Implementation follows a structured phased approach spanning approximately eight months from project kickoff to full production operation with validated disaster recovery capabilities. Each phase builds upon previous deliverables, reducing integration risk through incremental validation and creating natural decision points for go/no-go progression assessments. The timeline assumes availability of dedicated project resources including infrastructure engineers, security architects, network specialists, and operations personnel allocated sufficiently to execute parallel workstreams where the plan indicates concurrent activities.

6.1 Phase Overview

Phase	Duration	Focus Area	Key Deliverables	Exit Criteria
Phase 1: Facility Prep	Weeks 1-4	Physical infrastructure readiness	Power, cooling, racking, cabling complete	Facility inspection pass, power test successful
Phase 2: Base Platform	Weeks 5-8	Core infrastructure services	OS deployed, Docker running, network validated	All nodes reachable, docker-compose starts
Phase 3: Data Layer	Weeks 9-12	Databases, messaging, search engine	PG, Kafka, ES clusters operational	Smoke tests pass, replication working
Phase 4: Security Tools	Weeks 13-18	15 open-source tools deployed & integrated	All tools receiving data, cross-tool workflows	Integration tests pass, alerts generating
Phase 5: App Layer	Weeks 19-21	Frontend, API, auth, workflow automation	Full UI accessible, auth enforced	UAT sign-off from pilot users
Phase 6: Testing/Hardening	Weeks 22-25	Load test, pen test, failure simulation	Validated performance, hardened config	All tests green, runbooks complete
Phase 7: DR Setup	Weeks 26-29	DR site deploy, replication setup	DR site operational, data syncing	DR drill successful, RTO/RPO met
Phase 8: Go-Live	Weeks 30-33	Production cutover, hypercare support	Live traffic flowing, ops team trained	7-day stability, handoff complete

Table 6.1: Implementation Phase Summary

6.2 Phase 1 Detail: Facility Preparation (Weeks 1-4)

- Complete hardware procurement and receive all server, storage, networking equipment at primary site
- Install equipment in designated rack positions according to rack elevation diagrams
- Terminate fiber optic cables for 100GbE/25GbE data connections with proper testing
- Terminate copper cabling for 10GbE/1GbE management and out-of-band networks
- Configure UPS systems and validate automatic transfer switch operation via simulated failover test
- Deploy base operating system (Rocky Linux 9.4 or Ubuntu 22.04 LTS) on all fourteen primary site servers
- Install container runtime (Docker CE 27.x or Podman 5.x) and configure daemon.json settings
- Establish VLAN trunking from switches to server NICs according to network segmentation design

- Configure bonded network interfaces (LACP 802.3ad) for all multi-homed server connections
- Validate network throughput meets specifications using iperf3 at expected frame sizes
- Configure centralized logging (rsyslog forwarding to log aggregator) and chrony NTP synchronization
- Document all asset details (serial numbers, MAC addresses, IP addresses, rack positions) in CMDB
- Conduct physical security audit verifying access control, CCTV coverage, environmental sensors

6.3 Phase 2-4 Detail: Platform and Tool Deployment (Weeks 5-18)

Phases two through four encompass the technical heavy lifting of platform deployment, progressing from bare metal through containerized service orchestration to fully integrated security tooling. Phase two establishes the foundation: operating system hardening (CIS benchmarks), container runtime configuration, network bridge creation for each VLAN, and validation of docker-compose orchestration successfully scheduling containers across the cluster. Phase three deploys the data layer services: PostgreSQL cluster with PgBouncer, Apache Kafka with ZooKeeper, Elasticsearch cluster with Kibana, and Redis cache instances. Phase three concludes with data pipeline smoke tests confirming event flow from simulated sources through Kafka into Elasticsearch with searchable indices. Phase four introduces the fifteen security tools in dependency order, validating integration points between adjacent tools and confirming data flows match architectural expectations.

Week	Deployment Focus	Tools/Components	Validation Checkpoint
9-10	SIEM Foundation	Wazuh Manager, Elasticsearch Master, Kibana	Agent registers, index creates, dashboard renders
10-11	Event Pipeline	Wazuh Indexer, Kafka topics, Log shippers	Events flow end-to-end, alerts generate
11-12	NSM Engines	Suricata IDS, Zeek/Bro NSM, File extraction	Rules fire, logs parse, files captured
12-13	Packet Analysis	Arkime/Moloch PCAP viewer, PCAP upload workflow	Full packet search works, sessions reconstruct
13-14	Endpoint (EDR)	GRR Rapid Response, Osquery Fleet server	Remote forensic access, endpoint queries
14-15	SOAR Platform	TheHive Case Management, Cortex Analyzers	Cases create, analyzers respond correctly
15-16	Threat Intelligence	MISP Threat Sharing, OpenCTI Platform	IOCs import, enrichment works, graphs render
16-17	Vulnerability	OpenVAS/GVM Scanner, DefectDojo VM	Scans launch, findings import, reports generate
17-18	Observability	Prometheus, Grafana, AlertManager, dashboards	Metrics collect, alerts fire, dashboards populate

Table 6.2: Security Tool Deployment Sequence

6.4 Phase 5-6 Detail: Application, Testing, Hardening (Weeks 19-25)

Phase five deploys the custom Next.js 16 frontend application and TypeScript backend APIs implementing telecom-specific business logic including fraud detection algorithms, CDR correlation engines, and subscriber profiling. Authentication framework activates during this phase integrating with corporate LDAP directory (or local database fallback for air-gapped deployments) enforcing role-based access control across all platform functions. User acceptance testing proceeds with pilot analyst groups providing feedback on workflow usability and feature completeness. Phase six subjects the integrated platform to rigorous testing: load tests sustaining fifty thousand EPS for minimum twenty-four hours, penetration testing by authorized red team assessing application and API security posture, failure mode testing simulating node failures and network partitions, and security configuration review validating TLS certificates, cipher suites, and header hardening meet organizational standards.

6.5 Phase 7-8 Detail: Disaster Recovery and Go-Live (Weeks 26-33)

Phase seven establishes the disaster recovery site infrastructure mirroring the primary site deployment at reduced scale. DR servers receive operating system deployment, container runtime installation, and platform service instantiation in standby configuration. Inter-site replication activates for all critical data stores with continuous monitoring of replication lag metrics. A formal DR drill exercises complete failover procedures validating that recovery time and point objectives are achievable under controlled conditions. Phase eight executes production cutover beginning with final data migration from legacy systems (if applicable), gradual traffic shifting to route live events through the new platform, and formal handoff from project team to operations organization. The hypercare period (weeks 32-33) maintains augmented staffing with developers on standby to rapidly address any issues discovered under genuine production load, concluding with transition to steady-state operations and standard change management processes.

7. Cost Estimation (100% On-Premises)

This section presents comprehensive cost estimates for deploying and operating the Djezzy SOC Platform entirely on-premises across two data center sites. Figures represent order-of-magnitude budgetary ranges based on enterprise-grade hardware list pricing from major vendors (Cisco, Dell EMC, HPE, Supermicro) and industry-standard professional services rates. Actual costs will vary based on vendor negotiations, geographic pricing variations, existing infrastructure leverage, and organization-specific scope adjustments. All costs are presented in United States Dollars (USD); organizations should convert to Algerian Dinar (DZD) using current exchange rates for local budget planning purposes.

7.1 Capital Expenditure - Primary Site

Category	Description	Unit Cost	Qty	Total (USD)
COMPUTE SERVERS				
SIEM Masters (x2)	Dual Xeon 6348, 512GB, 4TB NVMe	$42,000	2	$84,000
ES Data Nodes (x4)	Dual Xeon 6348, 512GB, 60TB NVMe	$58,000	4	$232,000
Database Cluster (x3)	Dual Xeon 4314, 384GB, 15TB NVMe	$38,000	3	$114,000
NSM Capture (x2)	Dual Xeon 6248R, 256GB, 20TB NVMe	$46,000	2	$92,000
Application (x2)	Single Xeon 6230, 128GB, 1TB NVMe	$16,000	2	$32,000
Archive Server (x1)	Dual Xeon 4310, 128GB, 384TB SAS	$35,000	1	$35,000
NETWORK INFRASTRUCTURE				
Spine Switches (x2)	100GbE modular chassis	$82,000	2	$164,000
Leaf Switches (x5)	25/100GbE fixed	$32,000	5	$160,000
Management Switches (x2)	10GbE L3 managed	$11,000	2	$22,000
Optics & Cabling	DAC, AOC, fiber, patch panels	-	1	$28,000
STORAGE EXPANSION				
NVMe Drives (spares)	7.68TB enterprise U.3	$1,700	8	$13,600
SAS HDD (archive)	16TB enterprise SAS	$420	24	$10,080
INFRASTRUCTURE				
UPS Systems	20kVA double-conversion	$14,000	2	$28,000
PDU Distribution	Switched monitored PDU	$3,200	8	$25,600
Rack Enclosures	42U closed perforated	$4,200	4	$16,800
PRIMARY SITE SUBTOTAL				$1,058,080

Table 7.1: Primary Site Capital Expenditure Breakdown

7.2 Capital Expenditure - Disaster Recovery Site

Category	Description	Unit Cost	Qty	Total (USD)
DR COMPUTE SERVERS				
DR SIEM/Analytics (x2)	Dual Xeon 5318Y, 256GB, 8TB NVMe	$28,000	2	$56,000
DR Database (x2)	Dual Xeon 4314, 192GB, 8TB NVMe	$26,000	2	$52,000
DR Application (x2)	Single Xeon 6230, 64GB, 1TB NVMe	$14,000	2	$28,000
DR Archive (x1)	Dual Xeon 4310, 96GB, 192TB SAS	$28,000	1	$28,000
DR Management (x1)	Single Xeon 4314, 32GB, 480GB SSD	$10,000	1	$10,000
DR NETWORKING				
DR Switches (x3)	25/100GbE leaf + mgmt	$24,000	3	$72,000
DR Optics/Cabling	Fiber, copper, adapters	-	1	$12,000
DR INFRASTRUCTURE				
UPS System	10kVA double-conversion	$9,000	1	$9,000
PDU Distribution	Switched PDU	$2,800	4	$11,200
Rack Enclosures	42U closed	$4,200	2	$8,400
DISASTER RECOVERY SUBTOTAL				$286,600

Table 7.2: Disaster Recovery Site Capital Expenditure

7.3 Software and Professional Services

Category	Description	Cost (USD)	Notes
Operating System Licenses	RHEL/Rocky Linux subscriptions (16 servers x 3yr)	$38,400	Or use Rocky free alternative
Container Runtime Support	Docker EE / Podman enterprise (optional)	$45,000	Community editions free
Backup Software	Enterprise backup suite (Veeam, Commvault)	$25,000	Perpetual + 3yr maintenance
Monitoring Suite	Datadog/New Relic or self-hosted Prometheus stack	$15,000	Self-hosted option near-zero cost
Deployment Services	Vendor/partner implementation assistance	$120,000	8-week engagement
Knowledge Transfer	Training program for ops team (certification)	$35,000	5-person cohort
Project Management	PM resources for 8-month duration	$80,000	1 FTE dedicated PM
Documentation	Technical documentation, runbooks, procedures	$25,000	Comprehensive docs package
Contingency Reserve	15% buffer for unforeseen costs	$218,450	Industry standard reserve
SOFTWARE & SERVICES TOTAL		$601,850	

Table 7.3: Software and Professional Services Costs

7.4 Total Investment Summary

Investment Category	Amount (USD)	Percentage
Primary Site Infrastructure	$1,058,080	53.1%
Disaster Recovery Site	$286,600	14.4%
Software & Licensing	$123,400	6.2%
Professional Services	$260,000	13.0%
Contingency Reserve (15%)	$218,450	11.0%
Contingency Reserve (15%)	$218,450	11.0%
TOTAL CAPITAL INVESTMENT	**$1,991,930**	**100%**

Table 7.4: Total Capital Investment Summary

7.5 Annual Operational Expenditure

Category	Annual Cost (USD)	Monthly Equivalent	Notes
Facility - Power (both sites)	$120,000	$10,000	~66kW avg @ $0.10/kWh, 24/7
Facility - Cooling (both sites)	$52,000	$4,333	Estimated 1.5x PUE factor
Facility - Colocation/Space	$72,000	$6,000	If not owned DC (6 racks x $1k/rack/mo)
Hardware Maintenance (12%)	$161,350	$13,446	4-hour response, parts included
Software Support Renewals	$41,133	$3,428	OS, backup, optional tools
Personnel - Platform Engineers (3 FTE)	$210,000	$17,500	$70k avg fully-loaded salary
Consumables - Media Replacement	$18,000	$1,500	Drive failures, growth
Training & Certification (annual)	$15,000	$1,250	Conferences, courses, certs
ANNUAL OPEX TOTAL	$689,483	$57,457	Excludes analyst personnel

Table 7.5: Estimated Annual Operational Expenditure

7.6 Five-Year Total Cost of Ownership

Cost Category	Year 1	Year 2	Year 3	Year 4	Year 5	5-Year Total
Capital Investment	$1,991,930	$0	$0	$0	$0	$1,991,930
Operational Expenditure	$689,483	$720,000	$750,000	$780,000	$810,000	$3,749,483

Hardware Refresh (Y3)	$0	$0	$400,000	$0	$0	$400,000
Major Upgrade Reserve	$0	$100,000	$0	$150,000	$0	$250,000
Annual Total	$2,681,413	$820,000	$1,150,000	$930,000	$810,000	$6,391,413
Cumulative Total	$2,681,413	$3,501,413	$4,651,413	$5,581,413	$6,391,413	-

Table 7.6: Five-Year Total Cost of Ownership Projection

The five-year total cost of ownership for the complete 100% on-premises Djezzy SOC Platform deployment approximates **6.4 million USD** (approximately 8.6 billion DZD at current exchange rates), representing an average annual investment of 1.28 million USD inclusive of personnel, facilities, maintenance, and periodic technology refresh cycles. This investment positions Djezzy with world-class security operations capabilities comparable to leading European and Middle Eastern telecommunications operators, while maintaining complete data sovereignty and operational independence from external service providers or cloud platforms.

8. Recommendations and Next Steps

8.1 Architecture Recommendation Confirmation

Based on your explicit requirement for 100% on-premises deployment with zero cloud dependencies, this guide presents a fully self-contained hardware architecture satisfying all functional, performance, and compliance requirements for the Djezzy National SOC Platform. The recommended configuration delivers enterprise-grade security operations capabilities including real-time event processing at fifty thousand events per second, line-rate packet capture at 100Gbps, multi-billion-event searchable repositories, integrated threat intelligence, automated incident response playbooks, and comprehensive vulnerability management. All infrastructure operates within Djezzy-owned facilities under complete organizational control, with air-gap capability ensuring continued operation without any external network connectivity once initial deployment completes.

8.2 Immediate Action Items (Next 30 Days)

Week 1: Facility Assessment

Conduct detailed assessment of primary data center site(s) evaluating available power capacity (kVA), cooling capacity (kW), rack floor space (square meters and RU count), network uplink availability (fiber providers, bandwidth options), and physical security posture (access control, CCTV, fire suppression). Document gaps requiring remediation before equipment installation.

Week 2: Vendor Engagement

Issue requests for quotation (RFQ) to preferred hardware vendors for server, storage, and networking equipment specified in this guide. Engage with at least three vendors to ensure competitive pricing. Request proof-of-concept evaluation units for critical components (particularly 100GbE capture NICs and NVMe storage arrays) to validate performance claims.

Week 3: Team Formation

Identify personnel who will form the core platform engineering team (target: 3-4 FTE for platform, 2 FTE for security operations). Assess skill gaps against required competencies (Linux administration, Docker/Kubernetes, Elasticsearch, networking, security tools). Initiate training plans for technologies new to the organization. Establish development/staging environment for practice deployments.

Week 4: Project Planning Finalization

Finalize detailed project schedule with resource loading, milestone dates, and risk register. Establish governance structure including steering committee, change control board, and escalation procedures. Secure executive sponsorship and budget authorization. Prepare procurement documentation for hardware purchase orders targeting Week 5-6 submission.

8.3 Success Metrics

Metric Category	Key Performance Indicator	Target	Measurement Method

Availability	Platform uptime (primary site)	> 99.99%	Prometheus/Grafana monitoring
Availability	DR failover time (RTO)	< 30 minutes	Quarterly DR drill timing
Performance	Events per second (sustained)	> 50,000 EPS	Kafka consumer metrics
Performance	Search query response (p95)	< 3 seconds	Kibana/Elasticsearch logging
Integrity	Data loss incidents	Zero	Replication lag monitoring
Security	Vulnerabilities (critical/high)	< 30 days remediation	OpenVAS/DefectDojo tracking
Operations	Runbook procedure coverage	> 90% documented	Documentation audit
Personnel	Team certification rate	> 80% certified	Training records review

Table 8.1: Post-Deployment Success Metrics Framework

This 100% On-Premises Hardware Architecture Guide provides a complete blueprint for deploying the Djezzy National SOC Platform entirely within Djezzy-owned infrastructure. The design satisfies the most stringent data sovereignty requirements while delivering capabilities comparable to or exceeding those of peer telecommunications operators globally. Successful execution of this plan will establish Djezzy as a regional leader in telecommunications security operations, with a platform capable of evolving alongside emerging threats and expanding operational scope as business requirements dictate. The air-gappable architecture ensures operational resilience regardless of external connectivity status, a critical capability for national infrastructure protection.